Examiner only

2. *(a)* When a mixture of iron(III) oxide and aluminium powder (Thermit mixture) is heated, there is a violent reaction. The reaction is carried out in a tube surrounded by a mound of sand because the temperature reaches 2500 °C. A bead of iron is recovered from the sand. The picture below shows the reaction taking place in a darkened room.

(i) Give the reason why the iron formed in the reaction is molten. [1]

....

....

(ii) Complete and balance the **symbol** equation for this reaction. [2]

$$Fe_2O_3 \ + \ 2Al \longrightarrow \boxed{}\,Fe \ + \ \ldots\ldots\ldots\ldots$$

(iii) State which of the substances is oxidised. Give the reason for your choice. [1]

....

....

(iv) When a mixture of magnesium oxide and aluminium powder is heated, there is no reaction.

List iron, magnesium and aluminium in order of reactivity. [1]

Most reactive ....

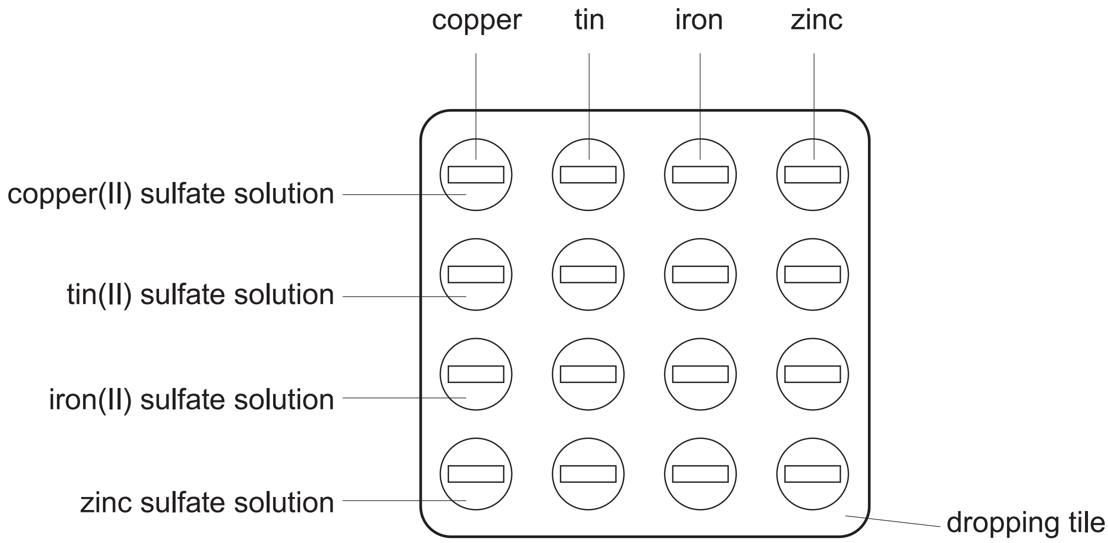
....

Least reactive ....

Examiner only

(b) Some metals are more reactive than others. A more reactive metal displaces a less reactive metal from its compounds.

A student was given tin, iron, copper and zinc and solutions of the metal sulfates. Using a dropping pipette, she put a little of one of the sulfate solutions in four of the depressions of the dropping tile. She did this for each solution in turn. She then put a piece of metal foil in each of the solutions, as shown below.



(i) Put a tick (✓) next to the question which **best** describes the investigation the student is carrying out. [1]

Which displacement is the most exothermic?	
Which metal can displace copper from solution?	
What is meant by the reactivity series?	
What are the positions of the four metals in the reactivity series?	

Examiner only

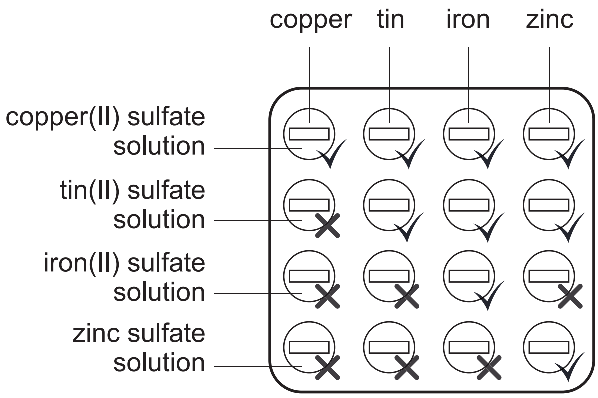
(ii) The student recorded the results by putting a tick (✓) next to a mixture which showed signs of a reaction and a cross (X) next to a mixture which showed no signs of a reaction.

The student concluded that:

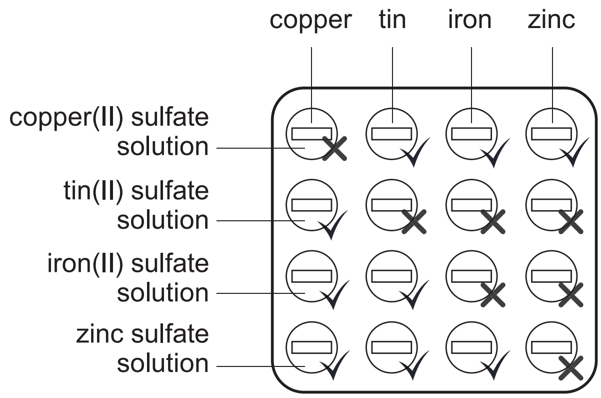
tin displaces copper
iron displaces tin
iron displaces copper
zinc displaces iron

Give the **letter** of the tile below which shows the results she recorded. [1]

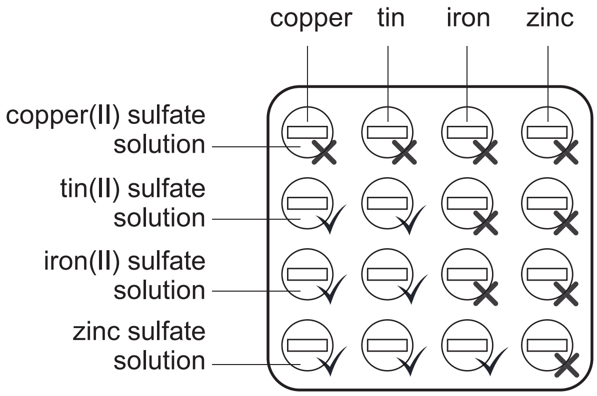
Letter ....



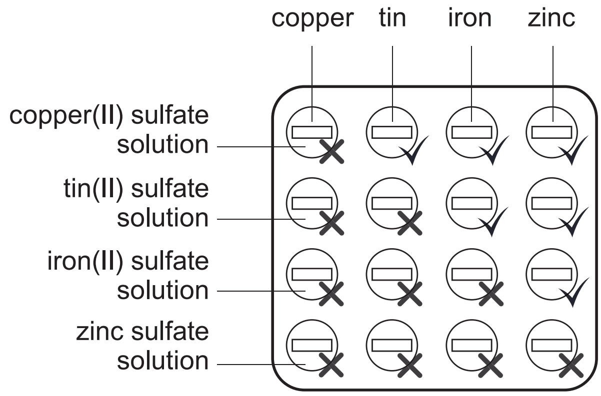
A



B



C



D

(iii) Another student said that not all of the tests were necessary. Give **one** example of a test not needed. Explain your choice. [2]

Example ....

Explanation ....

....

Examiner only

(c) Copper displaces silver from a solution of silver nitrate, $AgNO_3$, to form copper(II) nitrate solution.

(i) Describe **one** change the student would **see** during this displacement reaction. [1]

....

....

(ii) Write a balanced **symbol** equation for this reaction. [2]

....

12

3410UB01
07